

CHINNABABULREDDY VARRA

DATA ENGINEER

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SUMMARY

Data Engineer with 3.5+ years of experience in analyzing, designing, developing, and implementing data solutions. Skilled in applying SDLC methodologies, including Agile and Waterfall, to effectively manage project lifecycles. Proficient in programming languages such as Python, SQL, and Scala, with practical expertise in Big Data technologies including Hadoop, Map Reduce, Hive, Apache Spark, and Pig for optimized data processing and analysis. Experienced in designing and automating ETL workflows using tools like Apache NiFi and Talend to enhance operational efficiency. Adept at leveraging cloud platforms such as AWS and Azure to build scalable, secure, and cost-effective data solutions.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Programming Language: Python, R, SQL, Scala

IDE's: PyCharm, Jupyter Notebook

Packages: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, TensorFlow, ggplot2

Databases: MySQL, SQL Server, PostgreSQL, Oracle

Big Data Ecosystem: Hadoop, MapReduce, Hive, Pig, Apache Spark, Sqoop, Pyspark, Snowflake, HDFS

ETL Tools: SSIS, Apache NiFi, Apache Kafka, Talend, Apache Airflow, Informatica

Cloud Technologies: AWS, Azure, GCP, DataBricks

Reporting Tools: Tableau, Power BI, SSRS

Version Control Tools: Git, GitHub, GitLab

Other Skills: Data Cleaning, Data Wrangling, Critical Thinking, Communication & Presentation Skills, Problem-solving, Data Management

Operating Systems: Windows, Linux, Mac

EXPERIENCE

BANK OF AMERICA, USA | Data Engineer

Jan 2025 – PRESENT

- Collaborated with cross-functional Agile teams to define user stories and acceptance criteria, actively participating in sprint planning and backlog grooming, which ensured 100% on-time delivery of project milestones.
- Optimized data processing and automated ETL workflows using Talend, SSIS, and Hadoop ecosystem tools, resulting in a 30% increase in data flow efficiency while maintaining 99% data accuracy and integrity.
- Developed advanced data visualizations with Matplotlib and Seaborn, producing over 50 charts and plots that enhanced data communication and supported key business decision-making.
- Architected, deployed, and maintained scalable data pipelines on AWS leveraging services such as S3, EC2, Lambda, and Glue, reducing data processing time by 40% and improving overall pipeline reliability.
- Designed and delivered more than 30 custom reports using Tableau Desktop and SSRS, empowering stakeholders with actionable insights and improving data-driven decision-making by 25%.
- Managed cloud-based data architectures across AWS S3, Azure Data Lake, and Google Cloud BigQuery to enable scalable, secure, and cost-efficient analytics solutions.
- Automated data ingestion and transformation workflows with Apache Airflow, decreasing manual intervention by 50% and enhancing operational efficiency

CIGNA HEALTHCARE, USA | Data Engineer

Aug 2024 – Jan 2025

- Deployed Delta Lake over Azure Data Lake and AWS S3 to support version-controlled data storage, reducing redundancy and cutting storage costs by 30%.
- Enhanced data availability and optimized query performance for petabyte-scale healthcare datasets, supporting faster decision-making in clinical environments.
- Maintained strict adherence to HIPAA and healthcare data privacy standards, ensuring secure handling of sensitive patient information throughout the data lifecycle.
- Automated end-to-end CI/CD workflows for data pipelines using GitLab, Jenkins, and Terraform, cutting deployment times in half and improving reliability.
- Accelerated the delivery of real-time analytics to healthcare providers, minimizing delays in patient care decisions and improving operational responsiveness.
- Applied advanced data modeling strategies in Snowflake and BigQuery, reducing query execution time by 35% and boosting system efficiency for clinical reporting.
- Streamlined electronic health records (EHR) processing and clinical data retrieval, enabling clinicians to access critical insights more rapidly.
- Engineered scalable ETL pipelines using Azure Data Factory, Databricks, and PySpark, capable of processing over 5TB of healthcare data daily.
- Improved pipeline throughput by 40%, enabling real-time analytics integration with Azure Data Lake for faster, data-driven medical interventions.
- Facilitated seamless collaboration between data engineering, clinical, and data science teams to ensure alignment of infrastructure with hospital workflows and patient outcomes.
- Supported strategic healthcare decisions by delivering timely and actionable insights through efficient data pipeline management and robust cloud architecture.

Nefroverse Technologies, India | Jr. Data Engineer

Jun 2021 – Aug 2023

- Designed and optimized ETL pipelines using Python, SQL, and Hadoop, resulting in a 40% increase in data processing efficiency for structured healthcare datasets.
- Developed real-time data streaming architectures with Apache Kafka, incorporating schema validation to improve data quality by 30% for pharmacy inventory analytics.
- Led migration of on-premises SSIS workflows to AWS Glue and S3, reducing infrastructure costs and enabling scalable cloud-based data processing.
- Enhanced batch processing using Apache Spark on AWS EMR, cutting data pipeline execution time by 30% and improving accuracy of downstream analytics.
- Created automated deployment pipelines leveraging Jenkins and Docker, standardizing workflows and reducing deployment errors by 25%.
- Conducted ad hoc and large-scale data analysis with Amazon Athena and developed dynamic dashboards in Amazon QuickSight to visualize key healthcare KPIs, facilitating faster business decisions.
- Implemented agile methodologies to accelerate healthcare data initiatives, achieving a 20% reduction in delivery cycles through improved cross-functional collaboration.

EDUCATION

Master of Science in Information Technology and Management

Concordia University, Saint Paul

B.Tech in Computer Science and Engineering

Kalasalingam University of Research and Education

DATA SCIENCE 4 years course by IBM

PROJECTS

Responsive E-Commerce Web App

React, Redux Toolkit, TypeScript, REST API, CSS Modules

- Built a responsive e-commerce frontend with reusable components and dynamic routing.
- Integrated product APIs with authentication and cart functionality.

COVID-19 Support Platform for Small Businesses

React, Context API, HTML5, CSS3, Firebase

- Designed a user-friendly platform to connect small vendors with local customers.
- Enabled real-time product listing and secure payments.

CERTIFICATION

- AICTE-AWS Academy Cloud Foundation
- Data Analytics for Business – Coursera
- Computer Fundamentals – NPTEL